payment channel. Many cryptocurrency blockchains (including Bitcoin, Ethereum, and HyperLedger) have implemented the channel concept. The side channel concept is used in Bitcoin as a model for a lightning channel and in Ethereum as a state channel for off-chain transactions among trusted parties. Note that these channels are side channels besides the main cryptocurrency transfer channels. These side channels or side chains have been added to address scalability in blockchain networks, to improve transaction times, and to create micropayment channels.

> **DEFINITION** A *payment channel* is a means by which payments are transferred from one account to another. A *side channel* is an off-chain instrument enabled by the on-chain blockchain capabilities of smart contracts, hashing functions, cryptographic signatures, and identity management.

7.3 Micropayment channel

Micropayments are an age-old practice all over the world. Many local mom-and-pop economies depend on micropayments for daily living as well as for sustaining the local economy. These payments typically do not involve conventional financial institutions such as banks. With the advent of the digital age, efforts were focused on digitizing these micropayment methods, but they met limited success. The Bitcoin blockchain changed all that by proving the feasibility of online payments among unknown peers. With that breakthrough, interest in micropayments has been revived, and rightly so.

Here are some basic facts about a micropayment channel:

- It is defined by endpoints identified by the sender and receiver account addresses.